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Ad Campaign: Which Category to Feature

Introduction

As an aspiring creative director, I wanted to use this project as a chance to simulate a real-world marketing strategy using data to figure out how people shop at a particular online store and how that could influence ad campaigns. In today's digital economy, successful ads aren't created without consumer data backing the work; everything is strategized and carefully curated. I wanted to see how data science could guide the creative side of branding by predicting behavior and identifying patterns.

My focus was on analyzing shopping trends to uncover which types of products people gravitate toward, and what patterns could inform advertising decisions. I also wanted to include a section for the reviews of these products because without good reviews, products would not instill consumer loyalty. Customer loyalty is significantly important because that is the main way businesses gain profit, not through one time customers.

For this project, I worked with a shopping trends dataset that included customer demographics, purchasing behavior, and product feedback. I cleaned the data, visualized key insights such as ratings, trained a Random Forest Classifier to predict shopping habits, and created a visualization of the predicted number of buyers per category. This allowed me to get a better understanding of what the public would want to be advertised on a mass scale. Without personalizing advertisements for each potential consumer, a good clothing ad that could be sent out on a mass scale to lower costs yet still be effective.

While the clothing category had the highest number of purchases overall, one model predicting those purchases only reached about 52% accuracy. However, when I tested the other categories using Random Forest, the accuracy reached as high as 92%. Outerwear and Footware were not predicted to have a significant number of purchases showing: clothing was the most commonly purchased category. For someone interested in marketing strategy, this process showed how predictive modeling can back up creative decisions with grounded and objective insights.

Methods

The dataset I used was titled Shopping_Trends.csv and included data on:

- Gender
- Age
- Location
- Item Purchased
- Category
- Purchase Amount
- Review Rating

The dataset had no missing values and included both numerical and categorical variables, making it ideal for visualization and classification modeling.

Data Cleaning and Preprocessing

After importing the dataset, the data was already more or less clean. I looked for any missing values (which there were none) and found what categories were in the data to figure out what variables I was going to use. The column 'Category' was useful for visualizing consumer preferences, while 'Review Rating' gave insight into customer satisfaction.

I used label encoding to convert categorical features like Gender, Season, and Payment Method into numeric values. For modeling, I trained a separate Random Forest classifier for each product category using features like age, gender, review rating, and purchase history. The target variable in each case was a binary column indicating whether or not a user purchased that specific category.

Model Selection and Tools

To explore how shopping behavior varied across product categories, I trained a series of Random Forest Classifiers using scikit-learn. Each model was designed to predict whether a customer would purchase a specific category (like Clothing or Accessories) based on features such as age, gender, review rating, number of previous purchases, and total purchase amount. I used label encoding to convert categorical variables including Gender, Season, and Payment Method into numerical format so they could be used effectively in the model.

The dataset was split into training and testing sets (70/30 split), and a separate binary target variable was created for each category (e.g., bought_Clothing = 1 if a customer purchased clothing). Each model's performance was evaluated using a confusion matrix and overall accuracy score. Random Forest was chosen for its ability to handle mixed data types and its strength in managing feature importance without overfitting.

All analysis and visualization were completed in Python using pandas, seaborn, matplotlib, and scikit-learn.

Results

Figure 1: Purchase Counts by Category

A bar chart shows that 'Clothing' and 'Accessories' were the top 2 categories, with clothing being the most purchased overall.

Figure 2: Confusion Matrices by Category

A series of heatmaps displaying confusion matrices for four key product categories. Each matrix shows how the model predicted actual buyers versus non-buyers. Clothing had the most predicted purchases which makes sense knowing the results of the purchase counts by category.

Figure 3: Average Review Rating by Category

A bar chart that shows all product categories received generally high review ratings, with most falling above 3.5 out of 5. This suggests a strong trend of customer satisfaction across categories, rather than major differences between them.

Together, these results paint a picture of shopping trends across the dataset. Figure 1 confirmed that Clothing and Accessories were the most frequently purchased categories. This aligns with the Random Forest predictions in Figure 2; Clothing had the highest number of predicted buyers. This consistency between actual purchase counts and model output reinforces the reliability of the classifier. Figure 3 showed that review ratings were generally high across all categories, suggesting that customer satisfaction wasn't limited to a specific product type. Overall, the ad campaign should be focused around clothing.

Discussion

This project highlights how data can support curated marketing decisions by revealing behavioral patterns across shopping categories. The alignment between purchase frequency and model predictions suggests that machine learning can effectively mirror consumer behavior when supported by sufficient data. These insights help marketing teams decide which categories to prioritize in campaigns.

One of the biggest takeaways from this project was realizing how data can guide creative decisions in a meaningful way. As someone pursuing a career in marketing and creative direction, I usually think about campaigns from a storytelling perspective. Working with this dataset showed me how data science can find out what people care about and what motivates them to buy. Being able to pair that insight with storytelling would make campaigns more effective.

One of the biggest challenges I ran into was deciding which features to include in the Random Forest Classifier. There were so many options it wasn't obvious which variables would contribute the most to accurate predictions. I relied on trial and error, but in future projects, I'd use feature importance scores and more systematic testing to guide those decisions earlier.

If I were to expand on this project, I'd focus more on customer reviews as indicators of loyalty and long-term engagement. High review scores could signal repeat customers or brand trust. I'd look into whether customers who consistently leave high ratings are more likely to return or purchase across multiple categories. This insight could help businesses identify their most loyal customer segments and tailor campaigns to maintain those relationships.

Data science is a tool used every day by brands to personalize marketing at scale. I hope to keep building these technical skills so I can use them in future work that combines creativity with consumer understanding.

Appendix

Figure 1.

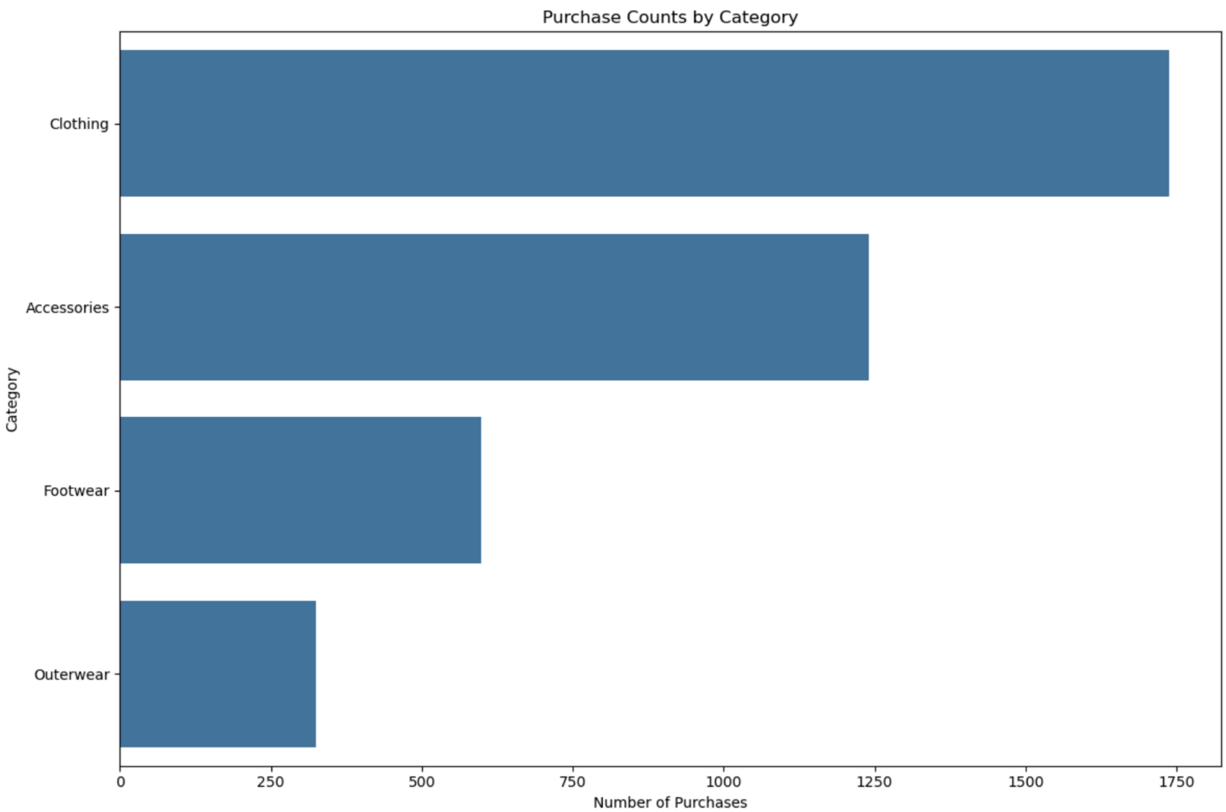


Figure 2.

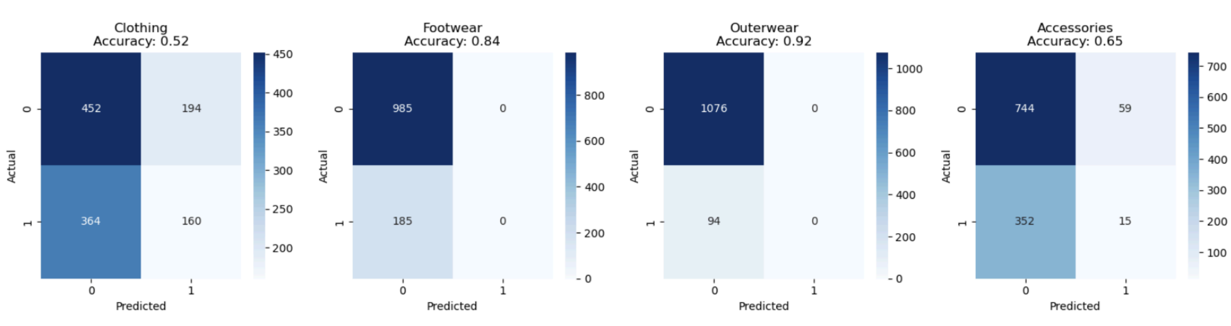
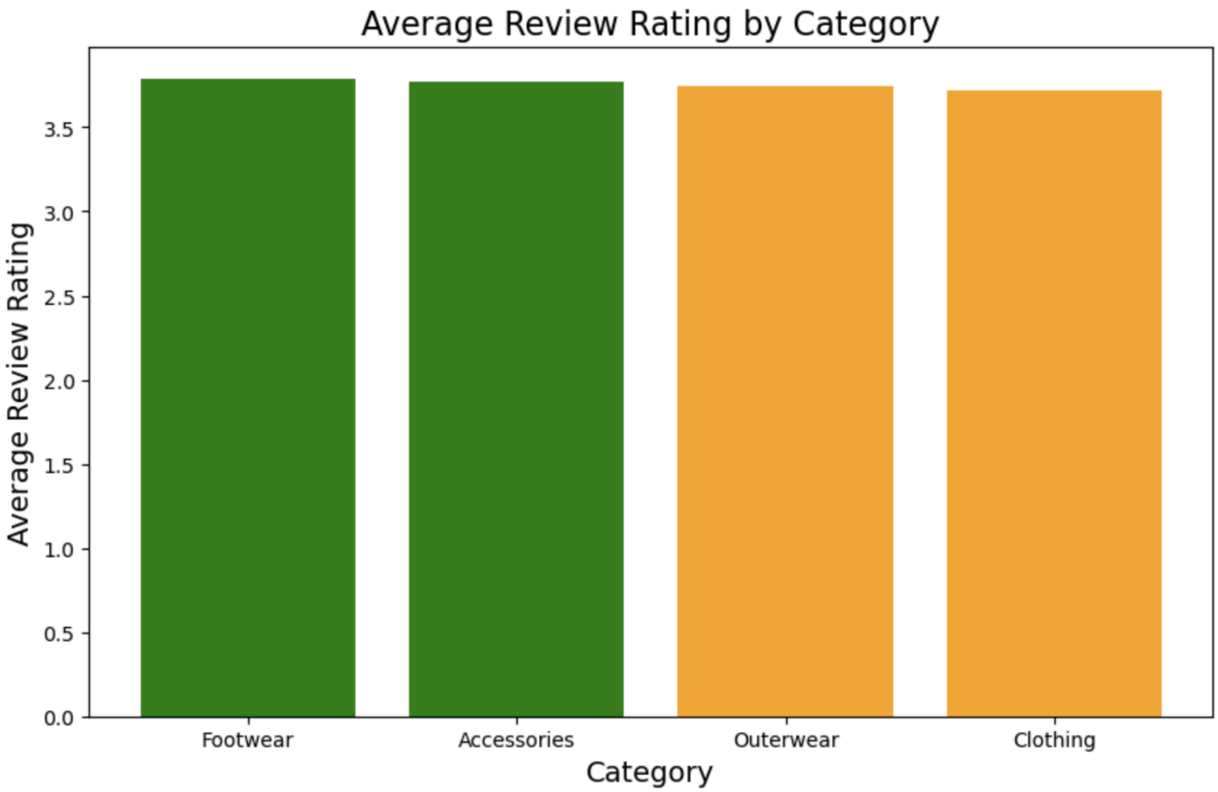


Figure 3.


Average Review Rating by Category
Average Review Rating
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Footwear
Accessories
Outerwear
Clothing
Category

Works Cited

heddgehogg. *Shopping Trends Dataset*. GitHub, 2022, https://github.com/heddgehogg/Shopping-Trends.

OpenAI. *ChatGPT by OpenAI*. Used to help with grammar and helped when I found errors in my code. My prompt was just the error and it would help me find a solution.